John Ogheneruese Fiewor

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Research Profile

AI engineer and founder working on trustworthy AI-assisted assessment for African classrooms. My work focuses on end-to-end assessment pipelines: paper-based exam digitisation, rubric-grounded grading, teacher-in-the-loop review, code-switched oral verification, offline/local execution for low-connectivity settings, and safety controls for multi-agent grading systems. I am especially interested in systems that preserve teacher authority while making assessment more scalable, auditable, and accessible.

Research Interests

- Artificial Intelligence in Education (AIEd)
- AI-assisted grading and feedback
- Teacher-in-the-loop and human-in-the-loop assessment systems
- Multilingual and code-switched educational NLP/ASR for African classrooms
- Multi-agent safety, validation, and auditability in grading pipelines
- Psychometrics, grading validity, and responsible educational AI
- Verifiable credentials and trustworthy educational achievement records

Education

Crawford University, Igbesa, Nigeria
M.Sc. Computer Science and Technology — GPA: 4.18 / 5.0
Jan 2021 – Oct 2022

Crawford University, Igbesa, Nigeria
B.Sc. Computer Science and Technology — Second Class Honours (Upper Division); GPA: 4.26 / 5.0
Nov 2014 – Jul 2018

Publications

Peer-Reviewed Workshop Papers

J. O. Fiewor. "Teacher-in-the-Loop AI Assessment for African Classrooms: Paper-Based Grading, Offline Exams, and Code-Switched Oral Verification." *NeurIPS 2026 Workshop on Global South AI*, 2026. Sole author. Accepted 23 Sept 2026. Submission 191. OpenReview: openreview.net/forum?id=IlKAI77Iwm (public release pending).

- Sole-authored study from GradrAI deployment work on paper-based grading under teacher review, code-switched oral assessment, and offline exam workflows.
- Reports a 60-script shadow-mode grading study; 42/60 and 43/60 scripts were routed to teacher review under two grading conditions.
- Evaluates code-switched classroom speech and African benchmark speech; language-hypothesis routing reduced mean WER from 0.63 to 0.15 and the downstream score gap from 2.9 to 1.3 marks out of 10.
- Includes an offline path for topic extraction, exam generation, and essay grading on teacher hardware using local model execution, schema-constrained decoding, recovery caching, and store-and-forward sync.

Research and Professional Experience

Founder & Lead AI Engineer, GradrAI — Lagos, Nigeria

Nov 2024 – Present | gradrai.com

- Founded and lead GradrAI, an AI-assisted assessment platform for automating paper-based exam grading workflows in higher-education settings.
- Architected the grading pipeline from raw document ingestion through text extraction, rubric interpretation, scoring, validation, teacher review routing, feedback generation, and analytics.
- Built the initial OCR-and-rule-based grading system using Microsoft Azure Cognitive Services, then led the transition to LLM-driven grading with Google Gemini on Vertex AI.
- Designed teacher-in-the-loop workflows so automated scores remain reviewable, auditable, and overrideable rather than replacing instructor judgment.
- Conducted shadow-mode evaluations on 60 paper-based scripts to compare grading conditions, identify invalid intermediate outputs, and measure the difference between final score validity and model self-confidence.
- Built code-switched oral-verification experiments for English mixed with Nigerian Pidgin, Yoruba, Swahili, and isiZulu, connecting speech recognition errors to downstream assessment effects.
- Developed an offline/local execution path for low-connectivity classrooms, using locally served models, schema-constrained decoding, recovery caching, and store-and-forward synchronization.
- Shipped the grading engine as a multi-agent system built with Google Agent Development Kit (ADK) and deployed on Vertex AI Agent Engine: sequential pipelines of specialised agents for preprocessing, rubric-based grading, referee cross-checking that routes low-confidence results to teacher review, weakness detection, personalised practice generation, and result aggregation, with MCP tools for database and storage access.

- **Technologies:** Google Gemini, Vertex AI, Google ADK, MCP, Microsoft Azure Cognitive Services, GCP, AWS, React, Node.js, Express, MongoDB, Mongoose.

Full-Stack Developer, VerifyEd — Hampshire, United Kingdom (Remote)

Feb 2025 – Present

Digital credentialing platform for issuing, rendering, sharing, and verifying educational credentials.

- Build and maintain issuer-facing and recipient-facing systems for educational credential issuance and verification across client and administrative interfaces.
- Shipped features improving credential portability and recipient control over presentation and sharing, including landscape-optimised certificate downloads and one-click sharing to professional networks.
- Contributed to the verification surface of educational achievement: how a credential is rendered, distributed, indexed, shared, and trusted by third parties once it leaves the issuing institution.
- Improved server-side rendering so credential pages are crawled more reliably and shared credential links appear correctly on LinkedIn and other platforms.
- Resolved production incidents affecting certificate verification, including XRPL ledger congestion and timeout scenarios.
- Designed and executed a custom script to recover unused XRP from dormant or invalid wallets, saving the company over $1,000.
- Built internal onboarding tooling for vetted customers and organisations, reducing fraudulent sign-ups and improving post-call onboarding flow.
- **Technologies:** Angular, Node.js, Express, PostgreSQL, Sequelize, AWS, Webflow.

Full-Stack Developer, Consolidated Hallmark Insurance — Lagos, Nigeria

Jan 2023 – Jun 2025

- Contributed to a unified internal API that automated core insurance business processes and improved interoperability across teams.
- Played a key role in insourcing the travel insurance portal, moving the product from third-party dependency to an internal platform with lower operational cost and faster iteration.
- Led backend and frontend enhancements that reduced the customer purchase journey from seven steps to three, increasing adoption and improving usability.
- Designed and implemented group, bulk, and Buy Now Pay Later purchase workflows for individuals, organisations, families, and sales agents.
- Temporarily led the frontend engineering team building a modern replacement for a legacy internal insurance operations application.
- Helped establish frontend architecture, CI/CD practices, codebase organisation, and team collaboration patterns.
- **Technologies:** React, JavaScript, Sass, Laravel, PHP, C#.NET, MySQL, SQL Server.

Junior Frontend Developer, HelloHub — London, United Kingdom (Remote)

Feb 2022 – Oct 2022

- Improved mobile responsiveness of the Admin Dashboard by translating Figma designs into production-ready React components.
- Wrote test-driven frontend code and refactored existing components to TypeScript, improving reliability and maintainability.
- Worked with component-driven development, React Testing Library, Jest, Storybook, and clean architecture practices.
- **Technologies:** React, TypeScript, JavaScript, Styled Components, React Testing Library, Jest, Storybook.js.

Frontend Web Development Intern, Zuri — Nigeria (Remote)

Mar 2021 – Jul 2021

- Developed a lease-management web application during a structured remote internship.
- **Technologies:** HTML, CSS, Gulp, Agile methodology.

Teaching and Mentoring Experience

Google Cloud Platform Mentor, Google Africa Developer Scholarship
Jun 2021 – Dec 2021

- Mentored learners working through Google Cloud Platform content and practical cloud-computing exercises.

High School Computer Science Teacher, Apostolic Faith Secondary School
Jan 2019 – Nov 2019

- Taught secondary-school computer science and supported students in foundational computing concepts and practical digital-skills development.

Selected Projects

GradrAI

gradrai.com

- End-to-end AI-assisted assessment platform for paper-based examinations, including document ingestion, grading, feedback, teacher review, analytics, and offline execution.

GradrAI Viva

github.com/Fiewor/gradrai-viva

- Open-source code-switched oral verification service and ASR benchmark (Deep Learning Indaba 2026, MLC Africa × Intron challenge): multi-hypothesis speech recognition routing, rubric grading, and WER/CER, downstream grade-gap, usable-transcript and latency evaluation on African-language speech.

MarkFlow

ican-grader.vercel.app

- Workflow-management platform for ICAN script marking, supporting examiner assignment, online manual marking, administrative review, and oversight.

Service

Reviewer, NeurIPS 2026 Workshop on Trustworthy AI for Good (AI4GOOD), 2026. Reviewed Submission 250, "From Design Specification to Manipulative UI: Auditing Specification-to-Behavior Failures in Agentic UI Generation." Reviewer recognition letter.

Awards and Honors

- **UNDP Nigeria and FSD Africa Innovation Challenge** — Winner, 2025, for work with RapidMedicare (rapidmed.care).
- **Google Gemini National #BuildWithAI Hackathon** — 3rd Place, 2024. Built the initial GradrAI MVP in three days using Google Gemini; independently implemented the frontend in under four hours after a teammate became unavailable and contributed the AI-driven backend component.
- **MongoDB World Hackathon** — 2nd Place, 600+ global participants, 2022. Co-developed "Good News," a web platform surfacing positive news stories from around the world.
- **IBM Call for Code for Racial Justice** — Open-source contributor and Hacktoberfest Demo Day speaker, 2021–2022. Contributed to the Truth Loop project, especially internationalisation and accessibility.

Certifications

- Python Coder — Kaggle, 2025–Present.
- Problem Solving (Basic) — HackerRank, 2025–Present.
- Machine Learning Specialization — DeepLearning.AI and Stanford University, 2024–Present.
- Microsoft Azure AI Engineer Associate, 2021–2024.
- Microsoft Azure Fundamentals, 2023–Present.
- IBM Carbon Design System Developer Essentials — React, 2021–Present.
- Google Cloud Certificate of Course Completion, 2021–Present.
- freeCodeCamp Responsive Web Design Certificate, 2020–Present.

Technical Skills

- **Languages:** JavaScript, TypeScript, Python, C#, HTML, CSS, Sass, SQL, Bash.
- **AI and Agentic Systems:** LLM-assisted grading, prompt engineering, rubric-grounded evaluation, multi-agent pipelines, MCP, Google ADK, Google Gemini, Vertex AI, Azure AI Services.
- **Frameworks and Libraries:** React, Next.js, Angular, Node.js, Express, Redux, .NET Core, Mongoose, Sequelize, Drizzle ORM.
- **Databases:** PostgreSQL, MongoDB, MySQL, SQL Server.
- **Cloud and DevOps:** Google Cloud Platform, AWS, Microsoft Azure, Git, CI/CD, REST APIs.
- **Testing and Engineering Practices:** Jest, React Testing Library, Storybook.js, Agile methodology, clean architecture.

Languages

- English — Native or bilingual proficiency.
- French — Limited working proficiency.

References

Available upon request.